

Task 5 — Enhancement research (one page)

Selected options: (a) smaller Docker images and (c) faster data processing.

Research. Docker documents separating build artifacts from runtime stages [1]. Pandas recommends built-in GroupBy operations over user-defined aggregation where possible [2]. These ideas were applied to Tasks 2 and 1/3 respectively.

Image change. The baseline uses the same Python 3.11 slim base, dependencies, source and dataset, with a conventional pip install that retains downloaded packages. The optimized Dockerfile builds wheels in a separate stage, mounts them only during installation, and disables the runtime pip cache. Neither compiler tools nor wheel archives are copied into runtime layers. Most of the measured saving is cache exclusion; a second stage alone does not guarantee a smaller Python image.

Processing change. Replace one DataFrame filter per diet per nutrient with one built-in groupby mean over all three nutrients. Batch analysis and the simulated function share this implementation. The benchmark asserts matching values and labels before timing; the baseline uses the same cleaned data.

Measurement	Before	After	Change
Image size (decimal MB)	697.66	568.16	18.56% smaller
Aggregation, 7,806 rows	13.722 ms	1.245 ms	11.02x faster
Aggregation, 156,120 rows	90.752 ms	11.964 ms	7.59x faster

Method. Image sizes are Docker image inspect Size values in decimal MB from the same VM and build base. Timings use perf_counter, one warmup and 15 measured calls per algorithm; medians are shown. The original dataset has 7,806 rows; 156,120 rows is an explicitly replicated 20x workload. Cleaning, file I/O, network download and plotting are excluded. Raw samples, resolved packages and image metadata are included in evidence/.

Expected benefit and limits. Excluding cache reduces image footprint; grouped aggregation avoids repeated scans. End-to-end deployment time and serverless cold starts were not measured. The timing improvements apply to this aggregation workload, and will vary by machine, data and dependencies. No cloud-cost savings are claimed.

[1] Docker, Multi-stage builds: <https://docs.docker.com/build/building/multi-stage/>

[2] Pandas, Group by: split-apply-combine: https://pandas.pydata.org/docs/user_guide/groupby.html

Sources consulted 2026-09-30.